

Project Architecture

Indian Infra Jobs - Recruitment Portal

Working Architecture of Indian Infra Jobs

1. Project Overview

Indian Infra Jobs is a full-stack recruitment portal built with React and Firebase.

It serves two user groups: Job Seekers (Users) and Recruiters (Managers/Admins).

The platform handles job posting, candidate management, AI-powered ranking, and interview scheduling.

Tech Stack

- Frontend: React 19 + TypeScript + Vite 7
- Styling: Tailwind CSS 4
- Authentication: Firebase Auth
- Primary Database: Cloud Firestore
- Secondary Database: Supabase (job applications)
- AI Engine: OpenAI API (GPT-3.5/GPT-4o)
- Email Service: Supabase Edge Functions + SMTP
- Hosting: Firebase Hosting

Key Features

- Role-based access (User, Recruiter, Manager, Admin)
- Job posting and application management
- Resume upload and parsing
- AI-powered candidate scoring and ranking
- Candidate clustering using embeddings
- Interview scheduling with email notifications
- Real-time notifications
- Analytics dashboard with charts

2. Application Flow

Entry Point

index.html !' main.tsx !' App.tsx (BrowserRouter)

All routes are defined in App.tsx using React Router v7.

Authentication Flow

- User visits / !' LoginPage component renders
- Login: Firebase Auth (email/password)
- Registration: Creates user in Firebase Auth + Firestore
- Role check: Firestore users collection !' role field
- Redirect: Manager/Recruiter/Admin !' /dashboard, User !' /home
- Password reset via Firebase sendPasswordResetEmail

Route Structure

- / !' Home (Login/Register)
- /dashboard !' Manager/Recruiter/Admin Dashboard
- /jobs !' User Job Listings
- /job/:jobId !' Job Detail Page
- /home !' User Profile
- /interview !' Interview Response Page
- /verify-details !' Verify Details Page
- /notifications !' Notifications Page
- /personalized-cv !' AI Personalized CV
- /training-module !' Training Module
- /career-assistance !' Career Assistance
- /matching-jobs !' AI Matched Jobs

3. Component Architecture

Layout Components

- Sidebar.tsx - Main navigation sidebar for dashboard
- UserHeader.tsx - Header for user-facing pages
- TabHeader.tsx - Tab navigation within pages
- NotificationBell.tsx - Notification indicator

Auth Components

- LoginPage.tsx - Login/Register/Forgot Password

Dashboard Tabs (Manager View)

- JobPostsTab - Create, edit, manage job listings
- UploadResumesTab - Bulk resume upload and parsing
- CandidatesTab - View candidates per job posting
- ShortlistedTab - Shortlisted candidate details
- InterviewsTab - Interview scheduling and responses
- SelectedCandidatesTab - Final selected candidates
- RecruitmentPipelineTab - Pipeline visualization
- AnalyticsDashboard - Charts and metrics
- NotificationsTab - System notifications
- AddMembersTab - Invite team members
- AssessmentsTab - Candidate assessments
- ProfileTab - User profile management

User Components

- UserProfile - User profile and job browsing
- JobsAndApplicationsView - Job listings + applications
- ProfileDetailsView - Profile details editor
- StandardResumeTemplate - Resume template

Recruitment Components

- RecruitmentFormModal - Create new job posting
- RecruitmentCard - Job posting card
- RecruitmentDetailView - Detailed job view
- FilterSidebar - Job filters
- ShareJobModal - Share job via link
- PremiumRequestModal - Request premium access

Resume Components

- ResumeUpload - File upload with drag & drop
- CandidateList - Candidate listing
- CandidateDetail - Candidate detail view
- AIScoringModal - AI scoring trigger
- InterviewInviteModal - Send interview invite
- ManualDetailsModal - Manual candidate entry

4. AI Pipeline

Candidate Scoring (aiService.ts)

- Input: Candidate profile + Job Description
- Model: OpenAI GPT-3.5-turbo (cost-effective)
- Output: JSON { score: 0-100, reasoning: string }
- Storage: Firestore users/{id}.rankings.{jobId}
- Skips if already scored (caching)

Batch Ranking (rankingService.ts)

- Fetches job from Firestore recruits collection
- Fetches applicants from Supabase job_applications
- Processes each candidate sequentially
- Model: GPT-4o for higher accuracy
- Updates Firestore with ranking scores

Candidate Clustering (clusteringService.ts)

- Generates embeddings: text-embedding-3-small
- Implements K-Means clustering algorithm
- Cosine distance for similarity
- Auto-determines k: $\sqrt{n}$, min 3, max 6
- Labels groups via GPT-4o-mini
- Stores in Firestore candidate_clusters

5. Services & Utilities

Core Services (src/lib/)

- firebase.ts - Firebase app, Auth, Firestore, Storage init
- supabase.ts - Supabase client initialization
- openai.ts - OpenAI client (browser-side with API key)
- aiService.ts - Single candidate scoring
- rankingService.ts - Batch candidate ranking
- clusteringService.ts - Embedding + K-Means clustering
- jobApplications.ts - CRUD for job applications
- notificationHelper.ts - Create in-app notifications
- test-connections.ts - Connection testing utility

Utilities (src/utls/)

- emailService.ts - Email sending helper

Types (src/types/)

- index.ts - TypeScript interfaces (User, Candidate, etc.)
- pdf-parse.d.ts - PDF parsing type definitions

UI Components (src/components/ui/)

- Popup.tsx - Toast notification system (Context-based)
- ErrorBoundary.tsx - React error boundary
- CustomDropdown.tsx - Custom select dropdown

6. Email & Notifications

Email System (Supabase Edge Functions)

- Location: supabase/functions/index.ts
- Runtime: Deno (Supabase Edge Function)
- SMTP: Configurable (Mailtrap/Gmail)
- Templates: HTML emails with inline styles
- Types: interview_invite, member_welcome

Interview Invitation Email

- Contains role, round type, interviewers
- Lists available interview dates
- Action link to confirm availability
- Decline link for withdrawal

Member Welcome Email

- Sent when manager/recruiter is invited
- Contains temporary credentials
- Login link to portal

In-App Notifications (notificationHelper.ts)

- Stored in Firestore notifications collection
- Types: interview_invite, verify_details,
- congratulations, manager_invite,
- premium_request, premium_approved, premium_rejected
- Each has userId, type, title, message, metadata
- Read/unread status tracking

7. Key Data Flows

Job Seeker Flow

- 1. Register/Login !' Firebase Auth
- 2. Browse Jobs !' /jobs (UserDashboard)
- 3. View Job Detail !' /job/:jobId
- 4. Apply !' job_applications collection
- 5. Receive Interview Invite !' /interview
- 6. Select Date !' Response saved
- 7. Verify Details !' /verify-details
- 8. View Matched Jobs !' /matching-jobs

Recruiter Flow

- 1. Login !' /dashboard
- 2. Create Job Posting !' recruits collection
- 3. Upload Resumes !' Parse & store candidates
- 4. View Candidates !' Filter by job
- 5. AI Scoring !' Score candidates (GPT)
- 6. Shortlist !' Update candidate status
- 7. Send Interview Invite !' Email + notification
- 8. Track Responses !' Interview tab
- 9. Select Candidates !' Final selection

AI Scoring Flow

- 1. Recruiter triggers scoring for a job
- 2. Fetch candidates from Firestore
- 3. For each candidate: construct prompt
- 4. Call OpenAI API (GPT-3.5/4o)
- 5. Parse JSON response (score + reasoning)
- 6. Update Firestore rankings map
- 7. Return updated candidates to UI

Clustering Flow

- 1. Select candidate pool
- 2. Generate text embeddings (OpenAI)
- 3. Run K-Means algorithm
- 4. Label clusters via GPT-4o-mini
- 5. Store cluster assignments
- 6. Display grouped candidates

8. Configuration & Deployment

Environment Variables

- VITE_OPENAI_API_KEY - OpenAI API key
- Firebase config - hardcoded in firebase.ts
- Supabase URL + Anon Key - hardcoded in supabase.ts
- SMTP credentials - in Supabase Edge Function

Build & Deploy

- Dev: npm run dev (Vite dev server)
- Build: npm run build (tsc + vite build)
- Output: dist/ folder
- Hosting: Firebase Hosting
- SPA rewrites: All routes !' index.html

Project Structure

- src/pages/ - Page components (12 files)
- src/components/ - Reusable components
 - auth/ - Login
 - layout/ - Sidebar, Headers
 - tabs/ - Dashboard tab content
- recruitment/ - Job posting components
- resume/ - Resume management
- user/ - User profile components
- interview/ - Interview components
- analytics/ - Analytics dashboard
- ui/ - Shared UI components
- src/lib/ - Services and utilities
- src/types/ - TypeScript definitions
- src/utls/ - Helper functions
- supabase/functions/ - Edge functions

Key Dependencies

- react, react-dom ^19.1.0
- react-router-dom ^7.6.3
- firebase ^11.10.0
- @supabase/supabase-js ^2.50.2
- openai ^4.40.1
- recharts ^3.6.0 (analytics charts)
- lucide-react ^0.525.0 (icons)
- react-hot-toast ^2.5.2 (notifications)
- pdfjs-dist ^3.11.174 (resume parsing)
- jspdf ^4.2.1 (PDF generation)